

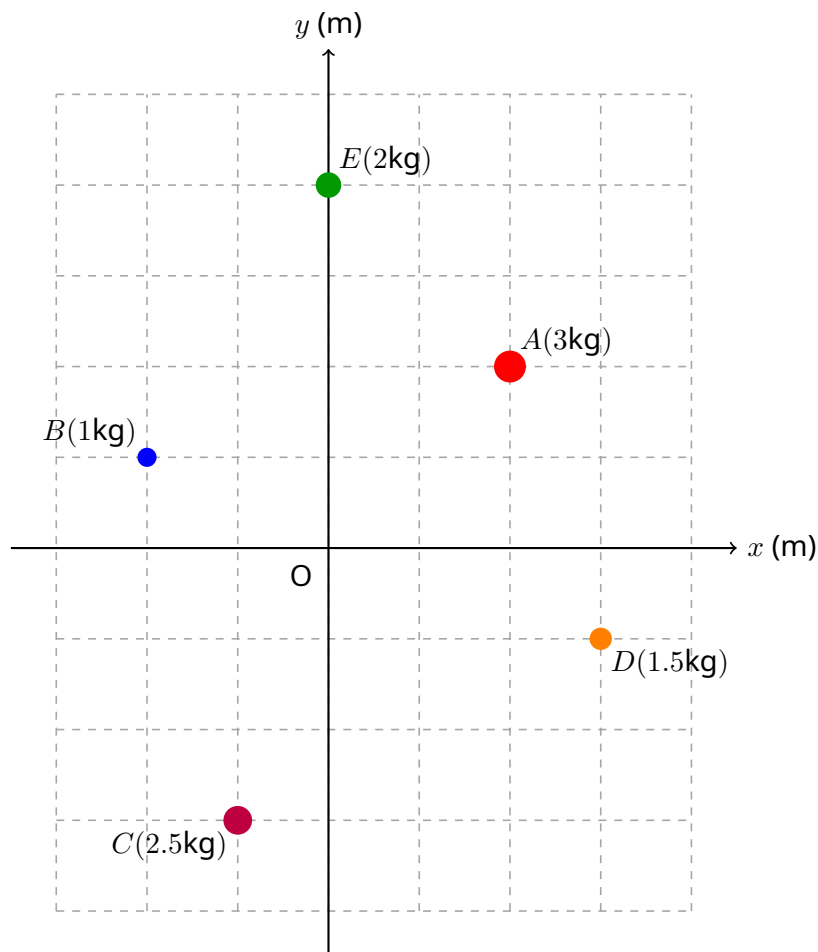
Supplementary Problem 2: System of Particles

Problem Statement

Consider a system of five particles distributed in a two-dimensional Cartesian coordinate system (measurements in meters). The masses and positions of the particles are given as follows:

- **Particle 1:** Mass $m_1 = 3$ kg located at point $A(2, 2)$.
- **Particle 2:** Mass $m_2 = 1$ kg located at point $B(-2, 1)$.
- **Particle 3:** Mass $m_3 = 2.5$ kg located at point $C(-1, -3)$.
- **Particle 4:** Mass $m_4 = 1.5$ kg located at point $D(3, -1)$.
- **Particle 5:** Mass $m_5 = 2$ kg located at point $E(0, 4)$.

System Diagram



Questions

1. Calculate the total mass M of the system.
2. Determine the abscissa x_G of the center of mass.
3. Determine the ordinate y_G of the center of mass.
4. Mark the approximate position of the center of mass G on the diagram above.